



RED-SERIES

PROFESSIONAL LOUDSPEAKER SYSTEM

RED-5 · RED-5.5 · RED-9 · RED-12 · RED-10

Complete Technical Reference & Installation Guide

DOCUMENT VERSION 1.0 · 2025

TABLE OF CONTENTS

1.	Introduction & Brand Overview	3
2.	Safety Instructions	4
3.	RED-5 Premium Professional Loudspeaker	5
3.1	Product Description	5
3.2	Key Highlights	5
3.3	Technical Specifications	5
3.4	Cabinet Dimensions	6
3.5	System Applications	6
4.	RED-5.5 Premium Professional Loudspeaker	7
4.1	Product Description	7
4.2	Key Highlights	7
4.3	Technical Specifications	7
4.4	Cabinet Dimensions	8
4.5	System Applications	8
5.	RED-9 Premium Professional Loudspeaker	9
5.1	Product Description	9
5.2	Key Highlights	9
5.3	Technical Specifications	9
5.4	Cabinet Dimensions	10
5.5	System Applications	10
6.	RED-12 Premium Professional Loudspeaker	11
6.1	Product Description	11
6.2	Key Highlights	11
6.3	Technical Specifications	11
6.4	Cabinet Dimensions	12
6.5	System Applications	12
7.	RED-10 Premium Professional Loudspeaker	13
7.1	Product Description	13
7.2	Key Highlights	13
7.3	Technical Specifications	13
7.4	Cabinet Dimensions	14
7.5	System Applications	14
8.	Series Comparison Table	15
9.	Installation & Setup Guidelines	16
10.	Amplifier Pairing Guide	17
11.	Warranty Information	18

1. Introduction & Brand Overview

Louis Martin Audio is an Italian professional audio brand dedicated to engineering loudspeaker systems that combine uncompromising acoustic performance with striking Italian aesthetics. Every product in the RED Series is conceived, engineered, and quality-tested to the highest professional standards, ensuring that installers and end users receive a system that performs flawlessly day after day.

The RED Series spans five models – RED-5, RED-5.5, RED-9, RED-12, and RED-10 – each sharing a common design philosophy: coaxial driver topology (except the RED-10 which uses a high-compression HF driver), passive crossover networks optimised for linearity, Glass Fiber Reinforced Plastic cabinets with premium car-paint glossy coating, and the distinctive aesthetic identity that makes Louis Martin products instantly recognisable in any installation environment.

This User Manual provides complete technical specifications, dimensional drawings, installation guidelines, amplifier pairing guidance, and warranty information for all four models. Read this document in full before installing or operating any RED Series loudspeaker.

Brand	Louis Martin Audio
Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Origin	Italy
Document	RED Series User Manual v1.0
Applies to	RED-5, RED-5.5, RED-9, RED-12, RED-10

2. Safety Instructions

■ **IMPORTANT – Read all safety instructions before installation or operation.**

Flying & Suspension	When suspending speakers overhead, approved rigging hardware rated for at least 10× the cabinet weight must be used. All flying apparatus must be inspected before every event by a qualified rigger. Secure all safety bonds/chains.
Regular Inspection	Inspect all speaker hardware, brackets, and flying apparatus before each use. Replace any component showing signs of corrosion, deformation, or wear immediately. Do not use damaged equipment.
Audience Distance	Maintain adequate distance between the loudspeaker and the nearest audience member. Consult local Health & Safety authority guidelines applicable to your region and venue type.
Digital Controller	The Louis Martin digital controller must only be programmed by an authorised technician. Operating any RED Series loudspeaker without the correct DSP preset voids the warranty and may cause permanent driver damage.
Environmental Protection	Protect all loudspeakers and associated electronics from humidity, water, rain, and condensation. These products are not rated for outdoor use unless additional weatherproofing is applied by a qualified installer.
Electrical Safety	All amplifiers and signal processors must be earthed. Use correctly rated cables and connectors. Do not exceed the rated power handling specified in this manual.
Hearing Safety	Prolonged exposure to high sound pressure levels causes permanent hearing damage. Ensure all personnel near operating loudspeakers wear appropriate hearing protection when levels may exceed 85 dB(A) for sustained periods.

RED-5

Coaxial Full-Range Passive Loudspeaker

3.1 Product Description

The RED-5 is a high-performance, versatile passive speaker system built around a single powerful 5.25" (133 mm) coaxial full range driver. Featuring full-range configuration, it delivers unmatched sound reinforcement for install market applications, including hospitality venues and premium home cinema. The ruler-flat response ensures transparent, uncoloured sound quality with exceptional vocal clarity and minimal distortion. The Glass Fiber Reinforced Plastic cabinet with car paint coating gives it a distinctive glossy finish equally at home in portable or fixed-installation scenarios.

3.2 Key Highlights

Key Highlights

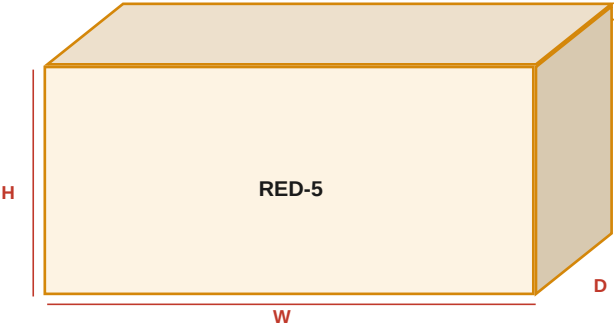
- ◆ Premium glossy Glass Fiber Reinforced Plastic cabinet
- ◆ 5.25" (133 mm) coaxial full range driver
- ◆ Passive crossover optimised for linearity
- ◆ 90°(H) × 90°(V) symmetric dispersion pattern
- ◆ Compact cube form-factor – 200 × 200 × 200 mm
- ◆ Multiple colour options available on demand

3.3 Technical Specifications

Parameter	Value
Type	Coaxial full-range passive loudspeaker
Frequency Response (–10 dB)	90 Hz – 19 kHz
LF Driver	1 × 5.25" (133 mm) Coaxial Full Range Driver
HF Driver	N/A
RMS Power	50 W
Program Power	100 W
Peak Power	200 W
Sensitivity (1W/1m)	93 dB
Max SPL Continuous (1m)	112 dB
Max SPL Peak (1m)	115 dB
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion	90°(H) × 90°(V)
Cabinet Material	Glass Fiber Reinforced Plastic, Car Paint Coating

Dimensions (H × W × D)	200 × 200 × 200 mm
Weight	6 kg / 13.2 lbs

3.4 Cabinet Dimensions & Diagram



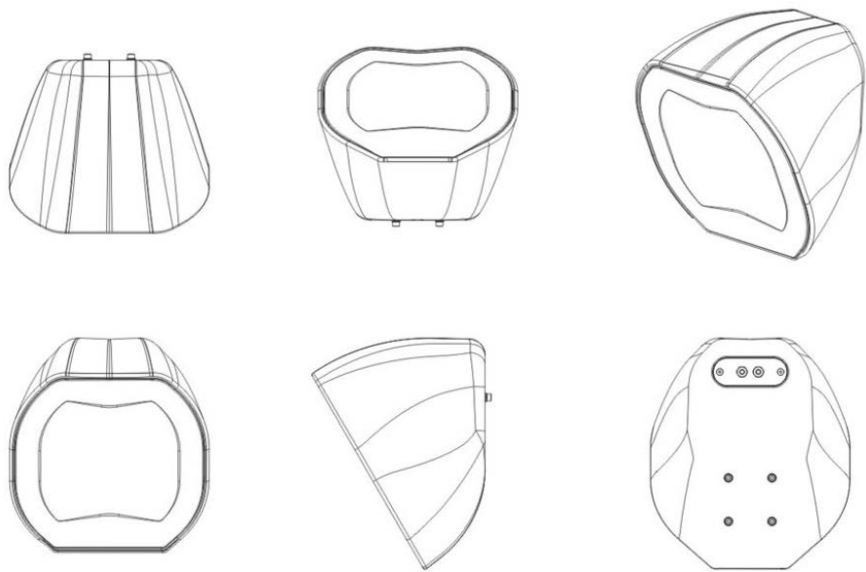
Dimension	Value
Height	200 mm
Width	200 mm
Depth	200 mm
Weight	6 kg (13.2 lbs)

All dimensions are nominal. H = Height, W = Width, D = Depth. Cabinet form factor: Cube / Spherical pod.

3.5 System Applications

Recommended Applications

- ◆ Lounge / Hospitality
- ◆ Pub / Bar
- ◆ Multi-functional room
- ◆ Delay / Fill speaker
- ◆ High-end Home Cinema



RED-5.5

Dual Coaxial Full-Range Passive Loudspeaker

4.1 Product Description

The RED-5.5 doubles the driver complement of the RED-5, housing two 5.25" (133 mm) coaxial drivers and two 1.34" (34 mm) HF drivers in a single compact enclosure. This dual-driver topology delivers 6 dB more sensitivity and significantly extended maximum SPL capability – ideal for nightclub environments and demanding home cinema installations where greater output is required without sacrificing the characteristic sound quality and transparency of the RED series. A wider 115°(H) horn provides excellent horizontal coverage for seated audiences and side-fill scenarios.

4.2 Key Highlights

Key Highlights

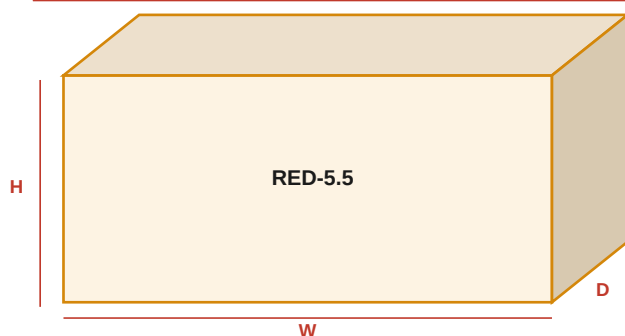
- ◆ Twin 5.25" (133 mm) coaxial drivers plus two 1.34" (34 mm) HF drivers
- ◆ Wide 115°(H) × 70°(V) coverage for broad audience areas
- ◆ 4 Ω impedance – compatible with high-current amplifiers
- ◆ Compact wide-body cabinet – 200 × 320 × 200 mm
- ◆ Premium glossy Glass Fiber Reinforced Plastic cabinet
- ◆ Multiple colour options available on demand

4.3 Technical Specifications

Parameter	Value
Type	Two-way Coaxial full-range passive loudspeaker
Frequency Response (–10 dB)	80 Hz – 19 kHz
LF Drivers	2 × 5.25" (133 mm) Coaxial Drivers
HF Drivers	2 × 1.34" (34 mm) HF Drivers
RMS Power	200 W
Program Power	400 W
Peak Power	800 W
Sensitivity (1W/1m)	96 dB
Max SPL Continuous (1m)	121 dB
Max SPL Peak (1m)	124 dB
Nominal Impedance	4 Ω (typical)
Crossover	Passive, optimised for linearity

Dispersion	115°(H) × 70°(V)
Cabinet Material	Glass Fiber Reinforced Plastic, Car Paint Coating
Dimensions (H × W × D)	200 × 320 × 200 mm
Weight	9 kg / 19.8 lbs

4.4 Cabinet Dimensions & Diagram



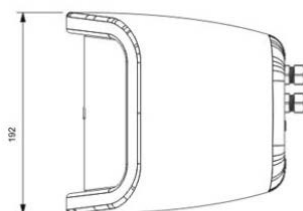
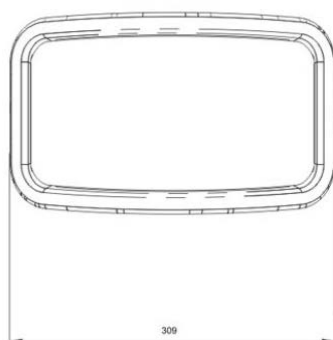
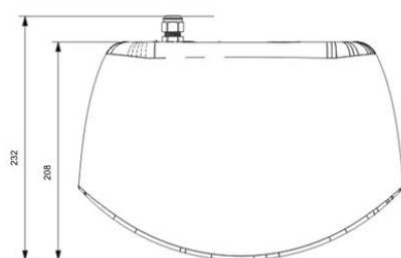
Dimension	Value
Height	200 mm
Width	320 mm
Depth	200 mm
Weight	9 kg (19.8 lbs)

All dimensions are nominal. H = Height, W = Width, D = Depth. Cabinet form factor: Wide horizontal pod.

4.5 System Applications

Recommended Applications

- ◆ Nightclub main system
- ◆ Pub / Bar fill
- ◆ Multi-functional room
- ◆ Delay / Fill speaker
- ◆ High-end Home Cinema



RED-9

10" Coaxial Full-Range Passive Loudspeaker

5.1 Product Description

The RED-9 scales the coaxial concept to a powerful 10-inch (254 mm) woofer paired with the same 1.34" (34 mm) HF driver used across the RED family. This step up in woofer diameter extends low-frequency reach to 75 Hz and increases sensitivity to 96 dB, matching the RED-5.5 in SPL capability but in a single, compact spherical pod. Engineered for nightclubs, restaurants, lounges, bars, and auditorium installations, the RED-9 maintains the signature transparent, ruler-flat response of the series with a focused 70°(H) × 70°(V) pattern for controlled coverage in reflective environments.

5.2 Key Highlights

Key Highlights

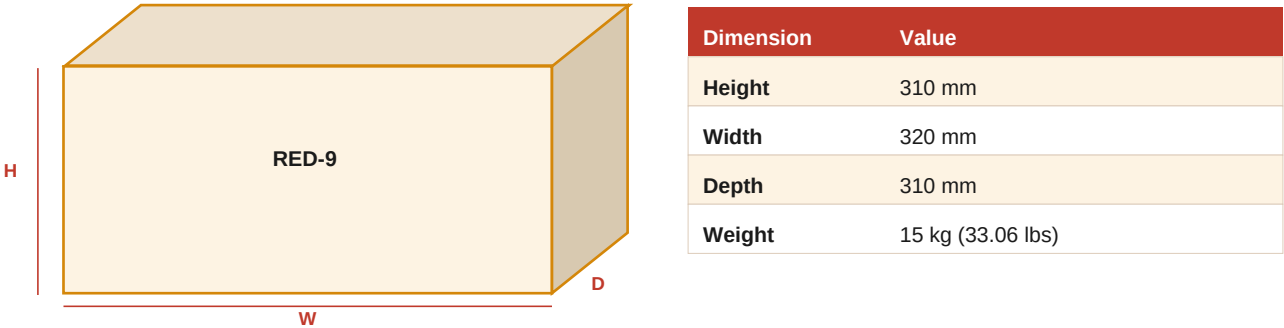
- ◆ 10" (254 mm) coaxial driver with 1.34" (34 mm) HF driver
- ◆ Focused 70°(H) × 70°(V) symmetric dispersion – ideal for reverberant spaces
- ◆ 8 Ω impedance – compatible with standard amplifiers
- ◆ Extended low-frequency response to 75 Hz
- ◆ Glass Fiber Reinforced Plastic spherical pod cabinet
- ◆ Multiple colour options available on demand

5.3 Technical Specifications

Parameter	Value
Type	Two-way Coaxial full-range passive loudspeaker
Frequency Response (–10 dB)	75 Hz – 19 kHz
LF Driver	1 × 10" (254 mm) Coaxial Driver
HF Driver	1 × 1.34" (34 mm) HF Driver
RMS Power	200 W
Program Power	400 W
Peak Power	800 W
Sensitivity (1W/1m)	96 dB
Max SPL Continuous (1m)	121 dB
Max SPL Peak (1m)	124 dB
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion	70°(H) × 70°(V)
Cabinet Material	Glass Fiber Reinforced Plastic, Car Paint Coating

Dimensions (H × W × D)	310 × 320 × 310 mm
Weight	15 kg / 33.06 lbs

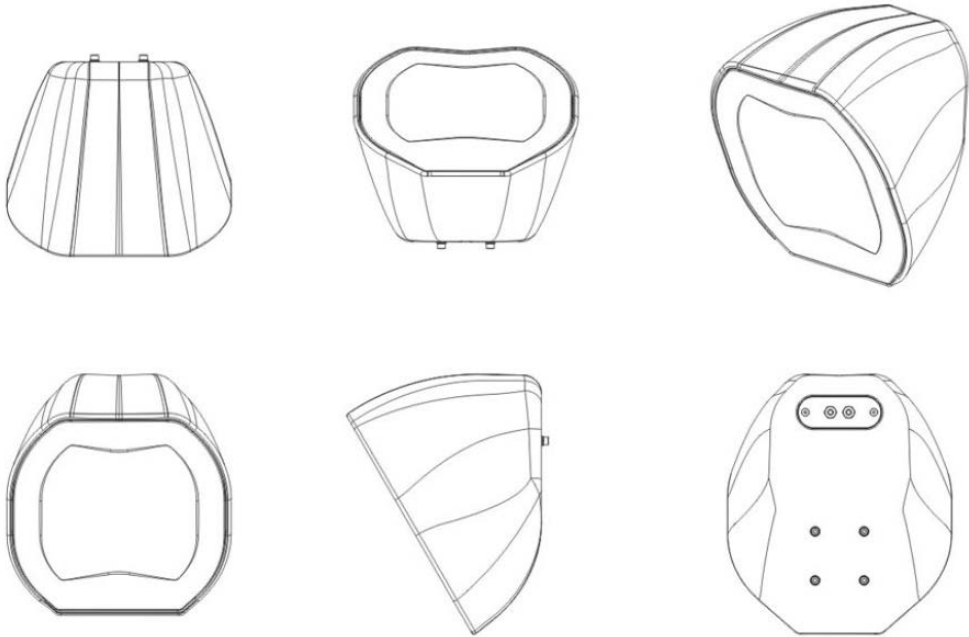
5.4 Cabinet Dimensions & Diagram



All dimensions are nominal. H = Height, W = Width, D = Depth. Cabinet form factor: Spherical pod.

5.5 System Applications

Recommended Applications
◆ Nightclub / Auditorium ◆ Restaurant fill ◆ Lounge / Pub / Bar ◆ Delay / Fill speaker ◆ High-end Home Cinema



RED-12

12" Coaxial Full-Range Passive Loudspeaker

6.1 Product Description

The RED-12 is a high-performance, versatile passive speaker system built around a powerful 12-inch (304 mm) coaxial driver paired with a 1.34-inch (34 mm) high-frequency driver. This two-way, full-range configuration combined with a passive crossover delivers exceptional audio quality across a wide range of installation environments. Engineered for nightclubs, restaurants, lounges, pubs, bars, and auditorium installations, the RED-12 provides a ruler-flat frequency response for transparent, uncoloured sound with outstanding vocal clarity, fidelity, and minimal distortion without compromising transient response. Constructed from Glass Fiber Reinforced Plastic with a premium car-paint glossy coating, the RED-12 is both durable and lightweight – equally suited to permanent fixed installations and portable use.

6.2 Key Highlights

Key Highlights

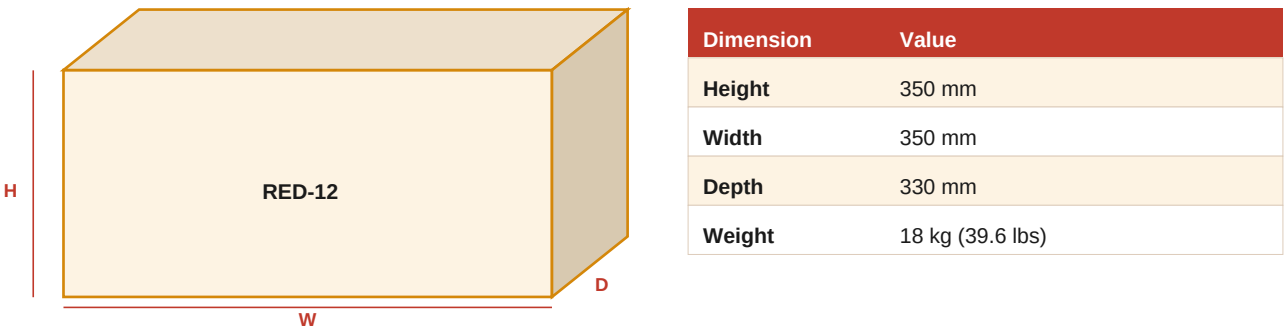
- ◆ Premium glossy Glass Fiber Reinforced Plastic cabinet
- ◆ 12" (304 mm) coaxial driver with 1.34" (34 mm) HF driver
- ◆ Passive crossover optimised for linearity
- ◆ 70°(H) × 70°(V) symmetric dispersion – controlled coverage in reverberant spaces
- ◆ Compact pod cabinet – 350 × 350 × 330 mm
- ◆ Multiple colour options available on demand

6.3 Technical Specifications

Parameter	Value
Type	Two-way Coaxial full-range passive loudspeaker
Frequency Response (–10 dB)	75 Hz – 19 kHz
LF Driver	1 × 12" (304 mm) Coaxial Driver
HF Driver	1 × 1.34" (34 mm) HF Driver
RMS Power	250 W
Program Power	500 W
Peak Power	1000 W
Sensitivity (1W/1m)	97 dB
Max SPL Continuous (1m)	123 dB
Max SPL Peak (1m)	126 dB
Nominal Impedance	8 Ω (typical)

Crossover	Passive, optimised for linearity
Dispersion	70°(H) × 70°(V)
Cabinet Material	Glass Fiber Reinforced Plastic, Car Paint Coating
Dimensions (H × W × D)	350 × 350 × 330 mm
Weight	18 kg / 39.6 lbs

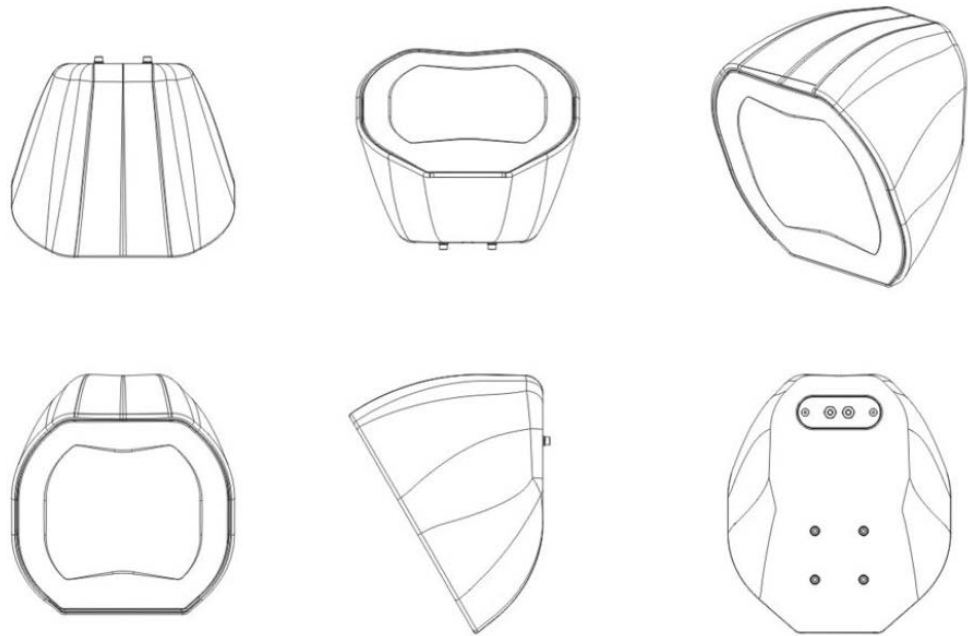
6.4 Cabinet Dimensions & Diagram



All dimensions are nominal. H = Height, W = Width, D = Depth. Cabinet form factor: Spherical pod.

6.5 System Applications

Recommended Applications
◆ Nightclub main / fill ◆ Lounge / Restaurant ◆ Pub / Bar ◆ Multi-functional room ◆ Delay / Fill speaker



RED-10

Dual 10" High-Power Full-Range Passive Loudspeaker

7.1 Product Description

The RED-10 is the flagship of the RED series – a high-power, dual 10" (254 mm) loudspeaker featuring a 1.75" (44 mm) high-compression driver for the HF section. With 600 W RMS continuous power handling and 131 dB continuous SPL, the RED-10 is engineered for the most demanding professional applications: nightclubs, large bars, private events, and permanent venue installations. The passive crossover ensures seamless integration between the two woofers and the compression driver, delivering the same ruler-flat, transparent response that defines the RED family, now at much greater scale. The distinctive binocular form factor is immediately recognisable and makes a bold visual statement in any installation.

7.2 Key Highlights

Key Highlights

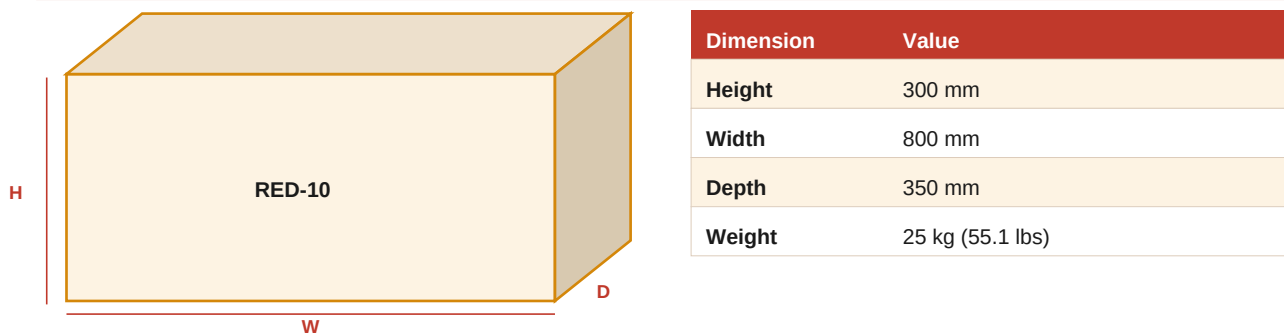
- ◆ 2 × 10" (254 mm) woofers with 2.5" (64 mm) voice coils
- ◆ 1.75" (44 mm) high-compression HF driver
- ◆ 600 W RMS / 1200 W Program / 2400 W Peak – flagship power handling
- ◆ 101 dB sensitivity – exceptional efficiency
- ◆ 90°(H) × 50°(V) wide-horizontal, controlled-vertical pattern
- ◆ Low-frequency extension to 65 Hz
- ◆ Distinctive binocular cabinet design with car paint glossy finish

7.3 Technical Specifications

Parameter	Value
Type	Two-way full-range passive loudspeaker
Frequency Response (-10 dB)	65 Hz – 18 kHz
LF Drivers	2 × 10" (254 mm) with 2.5" (64 mm) Voice Coil
HF Driver	1 × 1.75" (44 mm) High Compression Driver
RMS Power	600 W
Program Power	1200 W
Peak Power	2400 W
Sensitivity (1W/1m)	101 dB
Max SPL Continuous (1m)	131 dB
Max SPL Peak (1m)	134 dB

Nominal Impedance	4 Ω
Crossover	Passive
Dispersion	90°(H) × 50°(V)
Cabinet Material	Glass Fiber Reinforced Plastic, Car Paint Coating Glossy
Dimensions (H × W × D)	300 × 800 × 350 mm
Weight	25 kg / 55.1 lbs

7.4 Cabinet Dimensions & Diagram

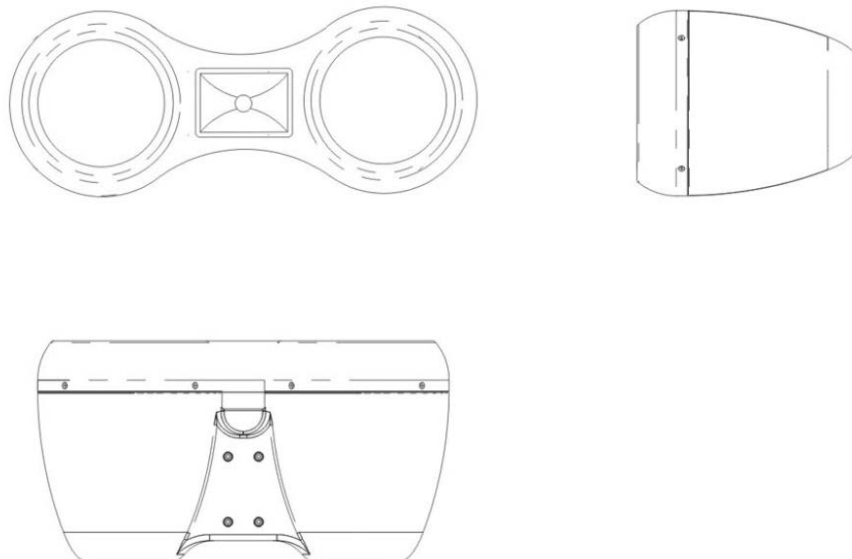


All dimensions are nominal. H = Height, W = Width, D = Depth. Cabinet form factor: Binocular / dual-pod.

7.5 System Applications

Recommended Applications

- ◆ Nightclub main system
- ◆ Large bar / pub
- ◆ Private events
- ◆ Permanent venue installation
- ◆ Stage fill / monitoring



8. Series Comparison Table

The table below provides a side-by-side comparison of all four RED Series models to assist in system design and model selection.

Parameter	RED-5	RED-5.5	RED-9	RED-12	RED-10
Frequency Response	90Hz–19kHz	80Hz–19kHz	75Hz–19kHz	75Hz–19kHz	65Hz–18kHz
LF Driver	1×5.25"	2×5.25"	1×10"	1×12"	2×10"
HF Driver	N/A	2×1.34"	1×1.34"	1×1.34"	1×1.75"
RMS Power	50W	200W	200W	250W	600W
Program Power	100W	400W	400W	500W	1200W
Peak Power	200W	800W	800W	1000W	2400W
Sensitivity	93dB	96dB	96dB	97dB	101dB
Max SPL (Cont.)	112dB	121dB	121dB	123dB	131dB
Max SPL (Peak)	115dB	124dB	124dB	126dB	134dB
Impedance	8Ω	4Ω	8Ω	8Ω	4Ω
Dispersion H×V	90°×90°	115°×70°	70°×70°	70°×70°	90°×50°
Dimensions (H×W×D)	200×200×200mm	200×320×200mm	310×320×310mm	350×350×330mm	300×800×350mm
Weight	6kg	9kg	15kg	18kg	25kg

9. Installation & Setup Guidelines

8.1 Pre-Installation Checklist

- Verify that all carton contents are present and undamaged.
- Check that the amplifier(s) meet the power and impedance requirements for the chosen model.
- Confirm the mounting hardware is rated for at least 10× the cabinet weight.
- Ensure the Louis Martin DSP controller is configured with the correct preset before powering on.
- Inspect all cables, connectors, and brackets before installation.

8.2 Mounting Options

- Wall-bracket mount: Use M10 or M12 stainless steel bolts through masonry or structural steel.
- Truss/flying mount: Use certified M10 eye bolts rated to EN 13411-3 or equivalent.
- Pole mount (RED-10): Use the integrated pole-mount socket with a 35 mm diameter pole.
- Safety bond: Always attach a secondary safety bond with a rated load of at least 500 kg.
- Allow free airspace around the cabinet for thermal management.

8.3 Cabling

- Use copper speaker cable with minimum 2.5 mm² cross-section for runs up to 10 m.
- For runs exceeding 10 m, increase to 4 mm² to minimise resistive loss.
- All connectors must be correctly rated Speakon NL4 or NL2 (Pin 1+/1-).
- Do not use unshielded cables in high-RF environments without ferrite suppression.

8.4 DSP Controller Setup

- Only an authorised Louis Martin technician may program the digital controller.
- Load the factory preset for the specific model before applying audio signal.
- Verify output levels are set to -10 dB before initial power-on.
- Gradually increase gain to operating level while monitoring for distortion.
- Save the preset configuration and label the controller channel clearly.

8.5 First Power-On

- Apply signal at very low level and listen for any unusual sounds.
- Slowly increase to nominal operating level over 30 seconds.
- Check all mounting hardware for vibration or resonance.
- Verify polarity with a battery test or phase measurement tool.
- Log the installation details including date, installer, and preset version.

10. Amplifier Pairing Guide

Selecting the correct amplifier is critical to obtaining optimum performance and reliability from RED Series loudspeakers. The guidelines below are based on program power (2× RMS) matching, which provides headroom for transients while protecting drivers from sustained clipping.

9.1 Power Matching Principles

- Match the amplifier rated output power at the speaker nominal impedance to 1×–2× the speaker RMS rating.
- Never use an amplifier whose clipped output exceeds the speaker program power.
- Always engage the amplifier limiter and high-pass filter as configured in the DSP preset.
- For 4 Ω loads (RED-5.5, RED-10), ensure the amplifier is rated at 4 Ω, not only 8 Ω.
- Bridged-mono operation doubles amplifier output and halves impedance seen – verify compatibility.

9.2 Recommended Amplifier Power Ranges

Model	Impedance	RMS Power	Min Amp Output	Max Amp Output	Notes
RED-5	8 Ω	50 W	50 W @ 8Ω	100 W @ 8Ω	Ideal for small zone amps
RED-5.5	4 Ω	200 W	200 W @ 4Ω	400 W @ 4Ω	Check amp 4Ω rating
RED-9	8 Ω	200 W	200 W @ 8Ω	400 W @ 8Ω	Standard power amp
RED-12	8 Ω	250 W	250 W @ 8Ω	500 W @ 8Ω	Standard power amp
RED-10	4 Ω	600 W	600 W @ 4Ω	1200 W @ 4Ω	High-current amplifier required

9.3 Impedance & Multi-Speaker Wiring

- Parallel wiring halves impedance: two 8 Ω speakers in parallel = 4 Ω.
- Series wiring doubles impedance: two 8 Ω speakers in series = 16 Ω.
- Never wire the RED-5.5 or RED-10 (already 4 Ω) in parallel – the resulting 2 Ω load will damage most amplifiers.
- For daisy-chaining multiple RED-5 or RED-9 units, calculate total impedance and verify amplifier stability.
- Always consult a qualified installer or contact Louis Martin Audio for multi-speaker system design assistance.

11. Warranty Information

Louis Martin Audio provides a comprehensive two-year limited warranty on all RED Series loudspeakers against defects in workmanship and materials from the original date of purchase.

Coverage Period	Two (2) years from the original date of purchase by the end user.
What is Covered	Manufacturing defects in materials or workmanship under normal use conditions. Louis Martin will, at its discretion, repair or replace the defective product at no charge for parts or labour.
What is NOT Covered	Damage resulting from: user abuse, accidents, unauthorised modifications, incorrect amplifier use, operation without the correct DSP preset, water or moisture ingress, physical impact, or failure to follow the safety and installation instructions in this manual.
DSP Controller	Any system failure directly attributable to incorrect programming of the Louis Martin digital controller by an unauthorised person is excluded from warranty coverage.
Warranty Claim Process	Contact your authorised Louis Martin distributor or email contact@louismartinaudio.it . The defective item must be adequately packed and dispatched pre-paid to the authorised service centre. Louis Martin will not accept freight-collect shipments.
Limitation of Liability	Louis Martin Audio's liability under this warranty is limited to repair or replacement of the defective product. Louis Martin is not liable for any indirect, incidental, or consequential damages arising from product failure.

Technical Support & Service

Louis Martin Audio | www.louismartinaudio.it

Email: contact@louismartinaudio.it

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